Appendix IV. Joint Fund-Bank Debt Sustainability Analysis

1. This debt sustainability is based on end-2005 data for external debt outstanding provided by the Direction de la Dette for most creditors and staff estimates for debts outstanding to IDA, the IMF, and the AfDB.[12] The overall staff assessment is that Niger is at a moderate risk of debt distress, although it remains vulnerable to a number of shocks.

2. **Following the Multilateral Debt Relief Initiative, Niger's debt outlook shows remarkable improvement over the short term, while trending upward during the medium and long term.**[13] The NPV of debt-to-GDP ratio declined from 22 percent in 2005 to 7.9 percent in 2006 as a result of the MDRI, while the NPV of debt-to-exports ratio fell from 136 percent in 2005 to 46 percent in 2006. However, Niger's high financing requirements, critical for promoting growth, will weigh heavily on the country's debt dynamics (Table 1b). Under the baseline scenario the NPV of debt-to-GDP ratio is expected to increase gradually to 22 percent in 2025, still lower than the policy-based indicative threshold (40 percent); and the NPV of debt-to-export ratio would rise gradually to 107 percent by 2025, significantly below the 150 percent threshold. The increase tapers off in the outer years and stabilizes close to 2021.

3. **Sensitivity tests show that Niger's debt burden would worsen, in the event of an adverse macroeconomic shock or weaker economic performance.** If key variables remain at the historical average of the previous eight years, the debt-to-exports ratio would rise to 120 percent in 2025. Also, a temporary reduction in export growth would have a pronounced impact on the country's debt indicators if the ensuing current account deficit had to be covered by borrowing.

4. **Favorable terms on new borrowing are critical to maintaining debt sustainability;** thus, if interest on new loans increase by 2 percentage points, the NPV of the debt-to-exports ratio would exceed modestly the indicative threshold, rising to 154 percent by 2025.

5. **The results from the DSA underscore the need for the authorities to pursue prudent debt policies combined with sound macroeconomic policies and reforms**. These should focus on growth-enhancing measures and investments, while continuing to seek grants and highly concessional loans in order to mitigate the likelihood of debt distress. The DSA suggests that although the risk of debt distress has declined with the granting of HIPC and MDRI assistance, debt management needs to remain very vigilant, given the country's continuing vulnerability to external shocks. This vulnerability must be reduced with sustained efforts to broaden the export base.

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[12] This analysis does not account for domestic debt, for which no reliable estimates exist and follows the granting of MDRI assistance in 2005.

[13] The indicative debt-burden thresholds for Niger are NPV of debt-to-exports ratio of 150 percent; NPV of debt-to-GDP of 40 percent, and debt service to exports ratio of 20 percent.

Box 1. Baseline scenario assumptions

The baseline and alternative scenarios for 2006–25 is based on the following assumptions, which are broadly consistent with those used in the HIPC Completion Point document:

- Over the length of the projection period, real GDP growth is expected to rise from its historical average of 2.7 percent to an average of 4.3 percent in 2006–09, and further to about 5 percent by 2025. This results from investments in irrigation and infrastructure, which would raise productivity in the agriculture sector. This improvement, combined with efforts to diversify the economy, including through the development of tourism, would reduce vulnerability to macroeconomic shocks.
- Average inflation, as measured by the implicit GDP deflator, would average at about 1.5 percent.
- The revenue to GDP ratio is projected to rise gradually from 10.4 percent in 2005 to 12.6 percent by 2008 and to about 17 percent by 2025, reflecting the gradual convergence of the tax revenue ratio to the 17 percent WAEMU norm.
- Public expenditures, currently at 19.0 percent of GDP, would rise sharply to 23.0 percent by 2009, reflecting the authorities' efforts to embark on major projects to promote growth. However, it will eventually fall to about 20 percent of GDP over the longer term, as the authorities make progress towards meeting the Millennium Development Goals (MDGs).
- Total exports of goods and services, which benefited from a temporary boost in gold exports in 2005, are projected to grow at an annual average rate of 7.2 percent over 2009–25 in value, spurred by the strengthening of uranium and non-mineral exports. Imports are projected to grow at an annual average of 6.7 percent in value, broadly in line with nominal GDP growth. These assumptions lead to an improvement in the non-interest current account deficit from an average of 10.3 percent of GDP in 2006–10 to about 8.1 percent of GDP by 2025.

The baseline scenario assumes that half of total external financing needs (including project assistance) will be covered by concessional loans with a 50 percent grant element, excluding the borrowings from the IMF, while the remaining financial requirements are met by grants and net-FDI. Two alternative scenarios are considered; the first is based on borrowing at average historical terms; the second assumes that borrowing is made at less favorable terms (interest rates higher by 2 percent).

Table 1a. Niger: External Debt Sustainability Framework, Following Mulatilateral Debt Relief Initiative, 2005-2025[1]
(In percent of GDP, unless otherwise indicated)

	Act.	His. Ave.[2]	St. Dev.[2]	Est.	Projections								
	2005			2006	2007	2008	2009	2010	2011	2006-11 Ave.	2016	2025	2012-25 Ave.
External debt (nominal)[1]	**51.2**			**15.3**	**17.9**	**21.3**	**24.5**	**27.8**	**30.7**		**43.1**	**55.3**	
o/w public and publicly guaranteed (PPG)	51.2			15.3	17.9	21.3	24.5	27.8	30.7		43.1	55.3	
Change in external debt	-6.7			-35.9	2.5	3.5	3.2	3.3	2.9		2.1	0.7	
Identified net debt-creating flows	-0.8			5.4	9.7	9.6	9.6	8.5	8.0		7.1	5.7	
Non-interest current account deficit[3]	**7.3**	**5.9**	**1.0**	**7.3**	**10.4**	**10.5**	**10.7**	**9.8**	**9.4**	**9.7**	**9.1**	**8.1**	**8.9**
Deficit in balance of goods and services	13.3			12.3	13.2	13.4	14.1	13.0	12.4		11.8	11.0	
Exports	18.2			17.3	18.8	19.2	19.0	19.3	19.4		19.4	20.5	
Imports	31.5			29.6	32.0	32.5	33.1	32.3	31.8		31.2	31.5	
Net current transfers (negative = inflow)	-6.3	-3.7	1.3	-5.0	-2.8	-2.9	-3.5	-3.3	-3.1	-3.4	-2.8	-2.9	-2.8
o/w official	-3.2			-2.8	-0.9	-0.9	-1.5	-1.2	-1.0	-1.4	-0.4	0.0	
Other current account flows (negative = net inflow)	0.3			0.1	0.0	0.1	0.0	0.0	0.1		0.1	0.0	
Net FDI (negative = inflow)	**-0.4**	**-0.6**	**0.4**	**-0.4**	**-0.3**	**-0.4**	**-0.4**	**-0.4**	**-0.4**	**-0.4**	**-0.4**	**-0.3**	**-0.4**
Endogenous debt dynamics[4]	**-7.7**			**-1.5**	**-0.4**	**-0.5**	**-0.7**	**-0.8**	**-1.0**		**-1.5**	**-2.1**	
Denominator: 1+g+r+gr	1.1			1.0	1.1	1.1	1.1	1.1	1.1		1.1	1.1	
Contribution from nominal interest rate	0.1			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.3	
Contribution from real GDP growth	-3.4			-1.7	-0.6	-0.7	-0.9	-1.1	-1.2		-1.8	-2.4	
Contribution from price and exchange rate changes	-4.3			…	…	…	…	…	…		…	…	
Residual (3-4)[5]	**-5.9**			**-41.3**	**-7.1**	**-6.1**	**-6.4**	**-5.2**	**-5.1**		**-5.1**	**-5.0**	
o/w exceptional financing	0.0			0.00	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt[6]	21.9			7.9	8.8	10.6	12.3	13.7	14.9		19.1	22.0	
In percent of exports	136.4			45.6	46.5	55.1	64.7	70.9	76.9		98.6	107.1	
NPV of PPG external debt	**21.9**			**7.9**	**8.8**	**10.6**	**12.3**	**13.7**	**14.9**		**19.1**	**22.0**	
In percent of exports	136.4			45.6	46.5	55.1	64.7	70.9	76.9		98.6	107.1	
Debt service-to-exports ratio (in percent)	9.3			227.1	2.3	2.2	2.4	2.5	2.6		2.8	3.1	
PPG debt service-to-exports ratio (in percent)	**9.3**			227.1	2.3	2.2	2.4	2.5	2.6		2.8	3.1	
Total gross financing need (billions of U.S. dollars)	0.3			1.6	0.4	0.4	0.5	0.5	0.5		0.6	1.1	
Non-interest current account deficit that stabilizes debt ratio	10.3			43.2	7.9	7.1	7.5	6.4	6.5		7.0	7.4	
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.8	2.7	3.6	3.4	4.1	4.3	4.5	4.6	4.7	4.3	4.7	4.7	4.7
GDP deflator in US dollar terms (change in percent)	8.1	3.9	9.2	-4.0	4.4	2.4	2.4	2.5	2.8	1.8	2.0	2.0	2.0
Effective interest rate (percent)[7]	0.2	0.7	0.3	0.4	1.4	1.1	1.1	1.0	0.9	1.0	0.6	0.6	0.7
Growth of exports of G&S (US dollar terms, in percent)	13.7	7.4	12.7	-5.4	23.8	8.6	6.0	9.3	7.9	8.4	7.3	7.2	7.2
Growth of imports of G&S (US dollar terms, in percent)	23.8	10.3	15.1	-6.8	23.1	8.5	8.9	4.8	5.6	7.4	6.5	7.0	6.7
Grant element of new public sector borrowing (in percent)	...	...	...	53.0	52.5	50.0	50.0	50.0	50.0	50.9	50.0	50.0	50.0
Memorandym item:													
Nominal GDP (billions of US dollars)	3.4			3.5	3.8	4.1	4.4	4.7	5.1		7.0	12.7	

Source: Staff simulations.

[1]Includes both public and private sector external debt.

[2]Historical averages and standard deviations are generally derived over the period since 1997, subject to data availability.

[3]For 2007, the figures include additional financing assurances to cover the residual financing gap.

[4]Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

[5]Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

[6]Assumes that NPV of private sector debt is equivalent to its face value.

[7]Current-year interest payments devided by previous period debt stock.

Table 1b. Niger: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt following Mulatilateral Debt Cancellation Initiative, 2006-25
(In percent)

	Est.	Projections											
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	**2016**	2021	2025
						NPV of debt-to-GDP ratio							
Baseline	7.9	8.8	10.6	12.3	13.7	14.9	16.1	17.1	17.9	18.6	**19.1**	21.1	22.0
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2007-25[1]	7.9	8.8	12.8	15.5	17.2	18.7	20.1	21.2	22.1	22.9	**23.3**	24.5	24.6
A2. New public sector loans on less favorable terms in 2007-25 [2]	7.9	10.9	13.4	15.6	17.5	19.0	20.6	21.9	23.0	23.9	**24.5**	27.1	31.7
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	7.9	9.3	11.9	13.9	15.6	16.9	18.4	19.5	20.5	21.3	**21.9**	24.2	25.4
B2. Export value growth at historical average minus one standard deviation in 2007-08[3]	7.9	10.6	15.2	16.8	18.0	19.1	20.2	21.0	21.7	22.2	**22.4**	23.0	23.2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	7.9	9.8	13.0	15.2	17.1	18.6	20.2	21.4	22.5	23.4	**24.0**	26.6	27.8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08[4]	7.9	10.7	13.9	15.6	17.0	18.2	19.4	20.3	21.1	21.7	**22.1**	23.3	23.8
B5. Combination of B1-B4 using one-half standard deviation shocks	7.9	13.8	21.9	23.8	25.3	26.5	27.8	28.8	29.7	30.2	**30.2**	30.2	30.0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007[5]	7.9	12.4	15.1	17.6	19.7	21.4	23.2	24.7	25.9	26.9	**27.7**	30.6	32.0
						NPV of debt-to-exports ratio							
Baseline	46	47	55	65	71	77	84	89	93	97	**99**	105	107
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2006-25[1]	46	47	67	82	89	96	105	110	115	118	**120**	122	120
A2. New public sector loans on less favorable terms in 2006-25[2]	46	58	70	82	90	98	108	114	120	124	**126**	135	154
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	46	45	53	63	69	75	83	88	92	95	**97**	104	107
B2. Export value growth at historical average minus one standard deviation in 2007-08[3]	46	71	114	127	135	142	152	158	163	166	**166**	165	163
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	46	45	53	63	69	75	83	88	92	95	**97**	104	107
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08[4]	46	54	69	78	84	89	97	101	105	107	**109**	110	111
B5. Combination of B1-B4 using one-half standard deviation shocks	46	79	124	136	142	148	157	162	167	169	**169**	163	158
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007[5]	46	45	53	63	69	75	83	87	91	95	**97**	103	106
						Debt service-to-export ratio							
Baseline	5.4	2.3	2.2	2.4	2.5	2.6	2.8	2.8	2.8	2.8	**2.8**	3.0	3.1
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2007-25[1,6]	5.4	3.3	2.8	2.8	2.9	3.1	3.5	4.4	5.0	6.0	**7.9**	7.2	7.9
A2. New public sector loans on less favorable terms in 2007-25[2]	5.4	3.1	2.3	2.0	2.3	2.4	2.7	3.0	3.2	3.5	**3.7**	5.1	6.0
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	5.4	3.1	2.5	2.5	2.4	2.6	3.0	3.7	4.2	4.9	**5.4**	5.8	6.1
B2. Export value growth at historical average minus one standard deviation in 2007-08[3]	5.4	4.0	4.0	4.2	4.1	4.3	4.8	5.9	6.7	8.2	**9.9**	9.7	9.9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	5.4	3.1	2.5	2.5	2.4	2.6	3.0	3.7	4.2	4.9	**5.4**	5.8	6.1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08[4]	5.4	3.1	2.7	2.7	2.6	2.8	3.1	3.9	4.4	5.4	**6.2**	6.3	6.6
B5. Combination of B1-B4 using one-half standard deviation shocks	5.4	3.8	3.4	3.3	3.3	3.5	4.3	5.4	6.4	7.3	**7.9**	7.9	8.0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007[5]	5.4	3.1	2.5	2.5	2.4	2.6	3.0	3.7	4.2	4.9	**5.4**	5.7	6.1
Memorandum item:													
Grant element assumed on residual financing (i.e., financing required above baseline)[7]	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	**50.0**	50.0	50.0

Source: Staff projections and simulations.

[1]Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), and non-debt creating flows.

[2]Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

[3]Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

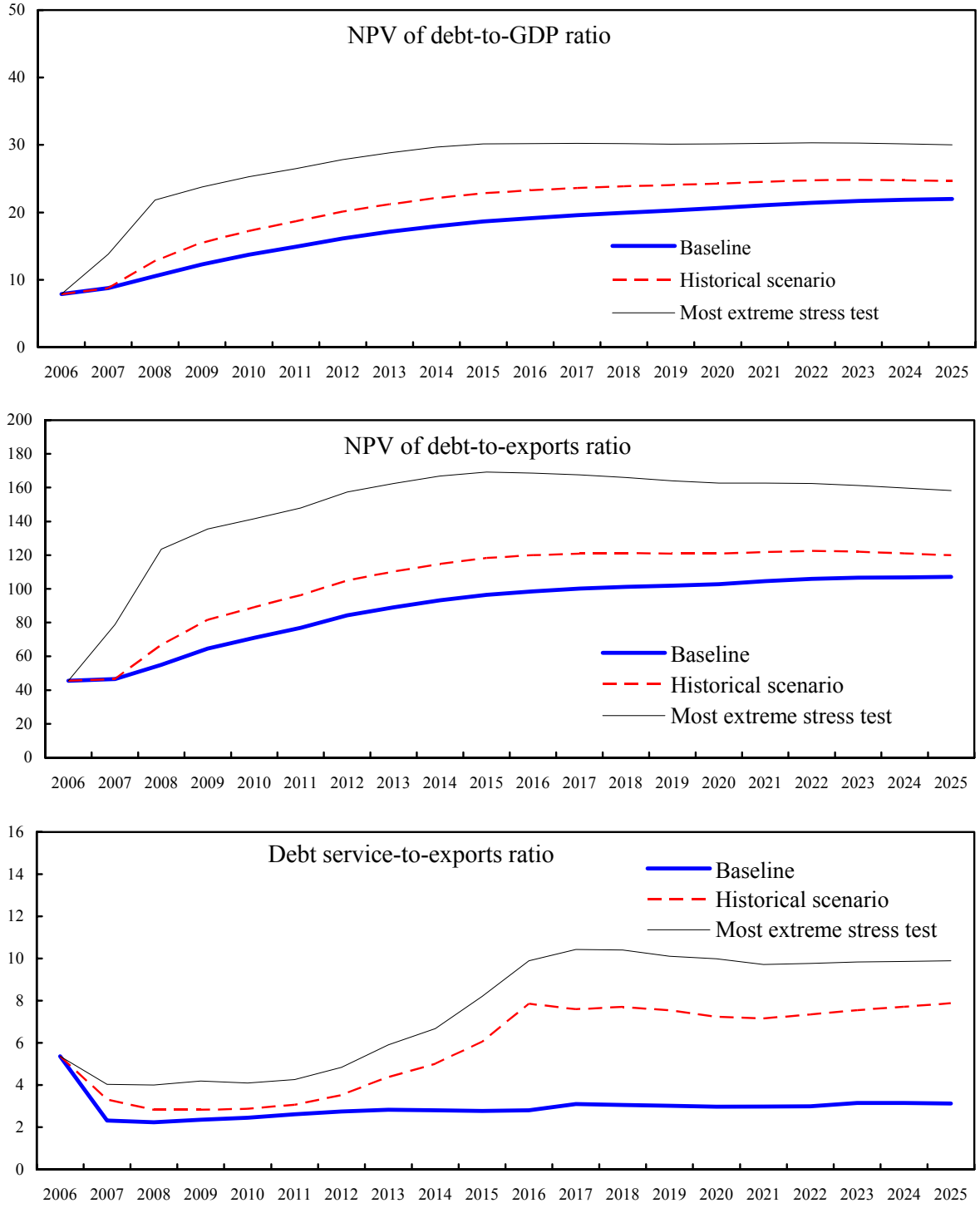
[4]Includes official and private transfers and FDI.

[5]Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6]In 2006, it excludes the impact of MDRI on debt service.

[7]Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Niger: Indicators of Public- and Publicly-Guaranteed External Debt
Following Mulatilateral Debt Relief Initiative, 2006-2025
(In percent)



Source: IMF staff projections and simulations.